

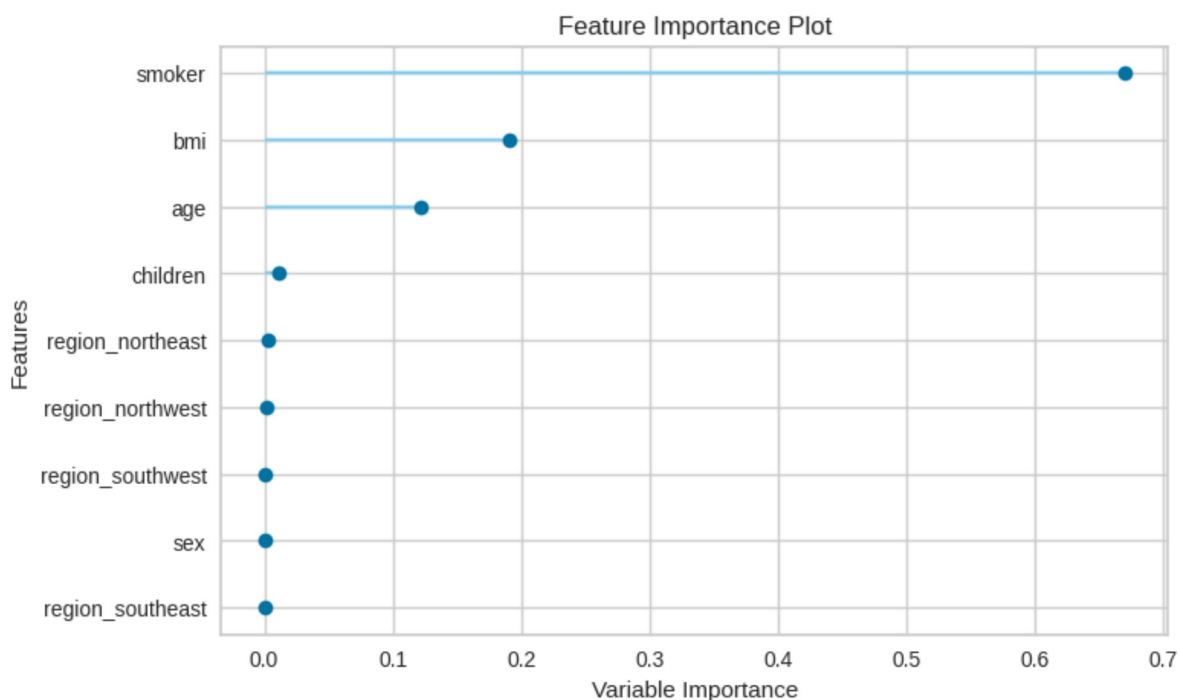
```
In [1]: from pycaret.regression import *
        from pycaret.datasets import get_data
        df = get_data('insurance')
        grid = setup(df, target='charges', session_id=42)
        best = compare_models()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

	Description	Value
0	Session id	42
1	Target	charges
2	Target type	Regression
3	Original data shape	(1338, 7)
4	Transformed data shape	(1338, 10)
5	Transformed train set shape	(936, 10)
6	Transformed test set shape	(402, 10)
7	Ordinal features	2
8	Numeric features	3
9	Categorical features	3
10	Preprocess	True
11	Imputation type	simple
12	Numeric imputation	mean
13	Categorical imputation	mode
14	Maximum one-hot encoding	25
15	Encoding method	None
16	Fold Generator	KFold
17	Fold Number	10
18	CPU Jobs	-1
19	Use GPU	False
20	Log Experiment	False
21	Experiment Name	reg-default-name
22	USI	d986

	Model	MAE	MSE	RMSE	R2	RMSLE	MAPE	TT (Sec)
gbr	Gradient Boosting Regressor	2610.8759	21791637.6860	4627.3067	0.8421	0.4277	0.3013	0.1970
lightgbm	Light Gradient Boosting Machine	2963.2674	25082834.4653	4972.9142	0.8191	0.5386	0.3615	0.1940
rf	Random Forest Regressor	2887.6516	25570645.6614	5029.7075	0.8169	0.4655	0.3337	0.2160
ada	AdaBoost Regressor	4165.9408	27820422.5003	5247.6376	0.8000	0.6048	0.6815	0.1800
et	Extra Trees Regressor	2872.2588	28630344.2699	5336.1453	0.7952	0.4755	0.3195	0.1990
ridge	Ridge Regression	4304.7348	38463769.8694	6185.4628	0.7247	0.6275	0.4370	0.1770
lar	Least Angle Regression	4293.1733	38465193.7000	6185.3822	0.7246	0.6125	0.4348	0.1860
llar	Lasso Least Angle Regression	4293.2896	38463709.9170	6185.2745	0.7246	0.6126	0.4349	0.1830
br	Bayesian Ridge	4299.5177	38464444.6848	6185.4300	0.7246	0.6188	0.4360	0.1880
lasso	Lasso Regression	4293.2897	38463694.1923	6185.2733	0.7246	0.6126	0.4349	0.2150
lr	Linear Regression	4293.1733	38465193.7000	6185.3822	0.7246	0.6125	0.4348	0.5820
dt	Decision Tree Regressor	3189.7676	44607432.0853	6641.9328	0.6854	0.5182	0.3346	0.1690
huber	Huber Regressor	3575.2076	48931102.8449	6957.4787	0.6479	0.4787	0.2251	0.1760
par	Passive Aggressive Regressor	4098.0528	61637269.9940	7813.9613	0.5715	0.4409	0.2496	0.1840
en	Elastic Net	7364.8412	90745151.7861	9496.6914	0.3691	0.7442	0.9431	0.1780
omp	Orthogonal Matching Pursuit	9140.1004	134861181.8891	11568.8040	0.0670	0.8889	1.1846	0.1710
knn	K Neighbors Regressor	8262.2066	136410752.8000	11625.8574	0.0505	0.8822	0.9978	0.1720
dummy	Dummy Regressor	9128.1113	146872878.4000	12071.9683	-0.0138	0.9997	1.5312	0.1860

```
In [2]: plot_model(best, plot='feature')
```



```
In [3]: # combining Regression, Time Series into one notebook
```

```
In [4]: from pycaret.datasets import get_data
from pycaret.time_series import *
df = get_data('airline')
grid = setup(df, fh=3, session_id=42)
best = compare_models()
plot_model(best, plot='forecast', data_kwargs={'fh' : 36})
```

```
Period
1949-01    112.0
1949-02    118.0
1949-03    132.0
1949-04    129.0
1949-05    121.0
Freq: M, Name: Number of airline passengers, dtype: float64
```

	Description	Value
0	session_id	42
1	Target	Number of airline passengers
2	Approach	Univariate
3	Exogenous Variables	Not Present
4	Original data shape	(144, 1)
5	Transformed data shape	(144, 1)
6	Transformed train set shape	(141, 1)
7	Transformed test set shape	(3, 1)
8	Rows with missing values	0.0%
9	Fold Generator	ExpandingWindowSplitter
10	Fold Number	3
11	Enforce Prediction Interval	False
12	Splits used for hyperparameters	all
13	User Defined Seasonal Period(s)	None
14	Ignore Seasonality Test	False
15	Seasonality Detection Algo	auto
16	Max Period to Consider	60
17	Seasonal Period(s) Tested	[12, 24, 36, 11, 48]
18	Significant Seasonal Period(s)	[12, 24, 36, 11, 48]
19	Significant Seasonal Period(s) without Harmonics	[48, 36, 11]
20	Remove Harmonics	False
21	Harmonics Order Method	harmonic_max
22	Num Seasonalities to Use	1
23	All Seasonalities to Use	[12]
24	Primary Seasonality	12
25	Seasonality Present	True
26	Target Strictly Positive	True
27	Target White Noise	No
28	Recommended d	1
29	Recommended Seasonal D	1
30	Preprocess	False
31	CPU Jobs	-1
32	Use GPU	False

	Description	Value
33	Log Experiment	False
34	Experiment Name	ts-default-name
35	USI	19d4

	Model	MASE	RMSSE	MAE	RMSE	MAPE	SMAPE	R2	TT (Sec)
ets	ETS	0.4912	0.5541	15.0940	19.3099	0.0318	0.0316	-0.4465	0.066
exp_smooth	Exponential Smoothing	0.4929	0.5560	15.1460	19.3779	0.0320	0.0317	-0.4600	0.066
arima	ARIMA	0.6964	0.7110	21.3757	24.7774	0.0447	0.0456	-0.5495	0.066
auto_arima	Auto ARIMA	0.7136	0.6945	21.9390	24.2138	0.0459	0.0464	-0.5454	8.606
lar_cds_dt	Least Angular Regressor w/ Cond. Deseasonalize & Detrending	0.8502	0.8264	26.2646	28.9923	0.0513	0.0534	0.0365	0.066
huber_cds_dt	Huber w/ Cond. Deseasonalize & Detrending	0.8658	0.8362	26.7826	29.3947	0.0516	0.0536	0.1501	0.066
lr_cds_dt	Linear w/ Cond. Deseasonalize & Detrending	0.8904	0.8722	27.5266	30.6243	0.0534	0.0555	-0.0092	0.076
ridge_cds_dt	Ridge w/ Cond. Deseasonalize & Detrending	0.8905	0.8722	27.5270	30.6246	0.0534	0.0555	-0.0092	0.086
en_cds_dt	Elastic Net w/ Cond. Deseasonalize & Detrending	0.8944	0.8746	27.6535	30.7127	0.0535	0.0557	-0.0063	0.086
lasso_cds_dt	Lasso w/ Cond. Deseasonalize & Detrending	0.8966	0.8759	27.7231	30.7594	0.0536	0.0558	-0.0040	0.076
llar_cds_dt	Lasso Least Angular Regressor w/ Cond. Deseasonalize & Detrending	0.8966	0.8758	27.7224	30.7580	0.0536	0.0558	-0.0042	0.056
br_cds_dt	Bayesian Ridge w/ Cond. Deseasonalize & Detrending	0.9156	0.8878	28.3188	31.1821	0.0547	0.0569	-0.0209	0.066
par_cds_dt	Passive Aggressive w/ Cond. Deseasonalize & Detrending	0.9652	0.9880	29.7382	34.5873	0.0619	0.0640	-0.7354	0.066

	Model	MASE	RMSSE	MAE	RMSE	MAPE	SMAPE	R2	TT (Sec)
knn_cds_dt	K Neighbors w/ Cond. Deseasonalize & Detrending	1.0695	0.9924	33.1500	34.9277	0.0631	0.0656	-0.1682	0.076
theta	Theta Forecaster	1.0839	1.0393	33.3223	36.2555	0.0686	0.0710	-1.7926	0.023
et_cds_dt	Extra Trees w/ Cond. Deseasonalize & Detrending	1.1865	1.1028	36.7229	38.7575	0.0708	0.0741	-0.5293	0.200
lightgbm_cds_dt	Light Gradient Boosting w/ Cond. Deseasonalize & Detrending	1.2019	1.1362	37.2359	39.9827	0.0713	0.0746	-0.6051	0.060
omp_cds_dt	Orthogonal Matching Pursuit w/ Cond. Deseasonalize & Detrending	1.2171	1.1475	37.6457	40.3070	0.0724	0.0757	-0.7057	0.070
dt_cds_dt	Decision Tree w/ Cond. Deseasonalize & Detrending	1.2784	1.1724	39.5420	41.1573	0.0779	0.0816	-1.0462	0.063
rf_cds_dt	Random Forest w/ Cond. Deseasonalize & Detrending	1.2784	1.1940	39.5319	41.9189	0.0764	0.0801	-0.9392	0.220
gbr_cds_dt	Gradient Boosting w/ Cond. Deseasonalize & Detrending	1.2849	1.1978	39.7617	42.0971	0.0768	0.0805	-0.8653	0.083
ada_cds_dt	AdaBoost w/ Cond. Deseasonalize & Detrending	1.3130	1.2455	40.6886	43.8232	0.0770	0.0812	-0.8082	0.100
naive	Naive Forecaster	1.5654	1.4951	48.4444	52.5232	0.0920	0.0981	-1.8344	0.153
snaive	Seasonal Naive Forecaster	1.6741	1.5343	51.6667	53.7350	0.1052	0.1117	-4.5388	0.120
polytrend	Polynomial Trend Forecaster	2.1553	2.1096	66.9817	74.4048	0.1241	0.1350	-4.2525	0.013

	Model	MASE	RMSSE	MAE	RMSE	MAPE	SMAPE	R2	TT (Sec)
croston	Croston	2.4565	2.3513	76.3953	82.9794	0.1394	0.1562	-4.5895	0.013
grand_means	Grand Means Forecaster	7.3065	6.5029	226.0502	228.3880	0.4469	0.5821	-72.1183	0.143

In [6]: *# there is supposed to be a plot above*